

BHAVESH GHANCHI

bhavesghanchi27@gmail.com | (323) 815-2582 | [linkedin.com/in/bhavesghanchi](https://www.linkedin.com/in/bhavesghanchi) | Los Angeles, CA

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, SQL
- **AI/ML:** PyTorch, TensorFlow, NLP, Transformers, LLMs, RAG, Fine-tuning (QLoRA), FAISS.
- **Backend & Systems:** Node.js, Express.js, FastAPI, RESTful APIs, MongoDB, PostgreSQL, MySQL, Distributed Systems.
- **Infrastructure & Tools:** Docker, Git, Linux, Jenkins, AWS, GCP

EDUCATION

University Of Southern California (USC), Los Angeles, CA August 2024 - May 2026
Master of Science in Computer Science (GPA 3.83/4)

Coursework: Analysis of Algorithms, Database Systems, Applied Natural Language Processing, Deep Learning

KJ Somaiya College of Engineering, Mumbai, India August 2020 - June 2024
Bachelor of Technology in Computer Engineering (CGPA 9.64/10)

PROFESSIONAL EXPERIENCE

FullStack Intern | React.js, TypeScript, Jest, Node.js, PostgreSQL, Express.js, Postman January 2024 - June 2024
Colgate-Palmolive Mumbai, India

- Built scalable frontend and backend features supporting **10,000+ users in Week 1 across 5 regions**.
- Optimized RESTful APIs and PostgreSQL queries, reducing **API latency by 25% under peak traffic**.
- Architected **CDN-backed multi-region architecture** and 4-language localization, increasing engagement by **35%**.
- Expanded Jest automated test coverage by **50%**, eliminating recurring production regressions.

Research Intern | Python, BeautifulSoup, PyPDF2, NLTK, NER, Pandas, NLP March 2023 - May 2023
K J Somaiya College of Engineering Mumbai, India

- Built a **multi-agent NLP pipeline** for clinical PDF ingestion, reducing manual preprocessing **effort by 60%**.
- Developed a **parallel-processing NER system** achieving **90%+ entity extraction** accuracy on biomedical corpora.
- Designed a **TF-IDF medical summarization** module that improved summary **precision by 25%**.

Backend Developer Intern | JavaScript, Node.js, MongoDB, Express.js, REST API, Firebase June 2022 - August 2022
SayHey Mumbai, India

- Owned backend services for **5,000+ users**, maintaining **99%+ uptime** and improving **API reliability**.
- Designed and optimized REST APIs handling **500+ concurrent sessions**, reducing booking latency by **30%**.
- Diagnosed root-cause issues in production, cutting **incident rate by 40%** and hotfix **turnaround by 35%**.

ACADEMIC PROJECTS

OTAS (Observability Tool for Agentic AI Systems) | Python, FastAPI, LLMs, Distributed Systems | [\[Github\]](#)

- Developed AI observability capturing agent decisions, prompts, tool calls, and latency, reducing debugging by **40%**.
- Built a **multi-agent AI platform** with **LLM orchestration**, **tool calling**, and workflow execution for **10+ agents**.
- Implemented **RBAC** and inference monitoring, reducing unnecessary **LLM API** calls by **30%**.
- Built a real-time dashboard for **800+ agent tasks/day**, achieving **96% workflow visibility** with **anomaly detection**.

Biomedical RAG Platform | Python, PyTorch, Hugging Face, FAISS, QLoRA | [\[HuggingFace\]](#)

- Built a RAG pipeline with **FAISS vector search** and embeddings, achieving **89% semantic retrieval accuracy**.
- Fine-tuned a **biomedical LLM** using **QLoRA**, improving answer quality by **+5.6% BERTScore F1** over baseline.
- Developed an **RAG-powered biomedical chatbot** delivering **84% relevant responses** for biomedical Q&A.

LEAP (Loss-aware End-to-end Acknowledged Protocol) | Java, UDP, Networking | [\[Github\]](#)

- Engineered a **TCP-like reliable transport** over UDP with sliding window, retransmission, and AIMD congestion control.
- Achieved **46 MB/s throughput** on 10 MiB transfers at 0% loss, benchmarked against a TCP baseline.
- Sustained **89% channel efficiency** across a 0–20% packet-loss sweep, validating robustness under adverse conditions.

Code2Text (Docstring Generator) | PyTorch, Transformers, FastAPI | [\[Github\]](#)

- Fine-tuned a **GPT-2 Transformer** on 160K+ Java functions, achieving **BERTScore F1: 0.84** (+18% over baseline).
- Optimized LLM inference through batching and optimization, **reducing latency by 35%** and **memory usage by 25%**.
- Containerized FastAPI model serving pipeline with **<500ms inference latency** for real-time IDE integration.